

COMPUTATIONAL PHILOLOGY · CORPORA · NEGATIVE RESULTS

Reading Steppe Names in Chinese Transcription

Three corpora, two medieval lexicons, and a measured answer to a fifty-year-old question about how much chance a sound law may tolerate.

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Preprint. Data and code: github.com/ulug2004/chinese-transcription-channel

ABSTRACT

Chinese histories record hundreds of Inner Asian names — Xiongnu rulers, Turkic titles, tribal designations — in characters chosen for their sound. Two thousand years of change in Chinese pronunciation have left the modern Mandarin readings of those characters unrelated to what the scribes heard. This paper models the transcription process itself as a noisy channel, trains it where both sides are known, and measures what can and cannot be recovered.

Four things result. **Three transcription corpora**, two of them extracted from sources not previously available as data: 9,329 Chinese–Mongolian pairs from the *Secret History of the Mongols*, 594 Chinese–Uyghur pairs from Ligeti's edition of a Ming glossary, and 1,017 verified Chinese–Sanskrit pairs separated out of a 27,110-entry Buddhist dictionary that mixes phonetic spellings with translations. **Two medieval Turkic lexicons** in machine-readable form, for which no downloadable dataset was found: 6,820 glossed headwords from the Türk Dil Kurumu index to Kāşgarî's *Dīwān Lughāt al-Turk* (1072–77), and 2,110 from Kuun's 1880 edition of the *Codex Cumanicus*. **A working channel model**, which on held-out data reconstructs 31% of Turkic words exactly and 84% within two segments. And **a set of measured negative results**: language attribution from transcription evidence is not supportable on available reference data, and full reconstruction of the Xiongnu names is not supportable at the density of evidence that survives — 1.4 observations per character, against the 60 that made the same method work on Mongolian.

The measurements also answer a question the field has carried unresolved since 1973. Doerfer argued that chance resemblance is pervasive in long-range comparison; his reviewer sharpened it into a threshold problem — *is it a sound law if 49% of cases show one correspondence and 51% another?* — and proposed three classes that nobody could then apply, for want of numbers. Section 7 applies them, with confidence intervals, and finds that the threshold cannot be a constant: it is the baseline of the component being measured, and it moves with how many outcomes that component has.

Keywords: historical phonology · noisy-channel models · corpus construction · Xiongnu · Old Turkic · null models · negative results

1 The problem

When a Chinese scribe of the Han dynasty wrote down a foreign name, he chose characters for their sound. 冒頓, the founder of the Xiongnu state, is read *Mòdùn* in modern Mandarin, but Mandarin is not the language the characters were chosen in. In the Later

Han period those two characters were pronounced approximately *mək tuən*.²¹ Every argument about what such a name meant, and about who the people bearing it were, depends on getting from the modern reading back to the period one.

That is a recoverable step: reconstructions of Later Han Chinese are published and are not in dispute here. The harder question is the one after it. Given a period reading, can the original foreign word be recovered? The transcription is lossy — Chinese had no *r*-, a restricted set of syllable shapes, and no way to write many foreign sounds — so the mapping from source word to spelling is many-to-one and its inverse is underdetermined. This is a noisy-channel problem, and it can be treated as one.

The interest of the Xiongnu case is that the answer is unknown and contested: proposals put the ruling house's language in the Turkic, Mongolic, Yeniseian and Iranian families, each resting on a handful of transcribed names. The interest of the *method* is that its reliability can be measured on cases where the answer is known, which is what most of this paper does.

2 A question the field has already asked

Nothing in the approach below is methodologically new, and it is worth saying so at the outset. Gerhard Doerfer made the argument in 1973, in *Lautgesetz und Zufall* — "Sound Law and Chance" — a study of what he called *omnicomparatism*: relating languages across great distances on the strength of resemblances.⁵ Chance produces such resemblances in quantity, was Doerfer's case, and comparativists have a repertoire of devices — he called them tricks — for making weak matches look strong. He included deliberately absurd comparisons of his own to show how easily the devices work.

A review in *Anthropos* the following year pressed the point into a question that can be answered with numbers.¹⁴ Sound laws, the reviewer observed, are not deterministic but probabilistic, and the literature never says how probabilistic they are permitted to be. *Is it a sound law if 49% of cases show the same correspondence and 51% show something else?* Where does regularity stop? He proposed three classes — deterministic laws, holding without exception; strong probabilistic laws, where one correspondence occurs in more than half of cases; and weak probabilistic laws, where one correspondence is merely the most common — and could apply none of them, for want of counts.

He also named the mathematics that was already available: the matching problem of Montmort (1708), which asks how often two shuffled sequences agree by accident. The permutation nulls used throughout this paper are that problem solved by simulation rather than by formula. What is new here is not the method but the measurement: the numbers are computed on this material instead of asserted, which is what Doerfer was asking for.

3 Resources

A learned model needs examples, and for this problem almost none existed as data. Five datasets were built: three of transcription pairs, and two lexicons of Turkic itself. All are released or reproducible; §12 sets out which is which and why.

3.1 Corpus 1 — *Secret History of the Mongols*

The *Secret History* is a thirteenth-century Mongolian chronicle preserved only in Chinese characters used phonetically, with an interlinear gloss. It is the largest known parallel text of its kind, and it had not been extracted as aligned pairs. Parsing the digital edition yields **9,329 Chinese–Mongolian transcription pairs**.

One detail decided whether the extraction worked. In the source encoding, a Chinese character followed by U+180B (a Mongolian free variation selector) marks an *annotation*, not part of the transcription. Treating those as data corrupts roughly a fifth of the corpus; the rule is one line, and finding it took considerably longer. Length agreement between the Chinese and Mongolian sides is 93.4%, which is the internal check that the alignment is real.

3.2 Corpus 2 — Ligeti's Sino-Uyghur vocabulary

Ligeti published a Ming-dynasty Chinese–Uyghur glossary in two instalments in *Acta Orientalia*.^{16,17} The Turkic forms are given in Latin transcription and the Chinese in EFEO romanisation, which makes the text parseable in principle and awkward in practice. Extraction yields **594 Turkic–Chinese pairs**.

Four properties of the typography had to be handled: the apostrophe is U+2019 rather than ASCII, Turkic forms may contain spaces, footnote digits intervene before the gloss, and the Turkic form is not always at the start of a line. A first pass that missed these returned 297 pairs, half of them wrong. A further trap was structural rather than typographic: journal running heads carry the author's name on verso pages, so a filter written for the article title alone silently discarded half the article.

3.3 Corpus 3 — verified Buddhist Sanskrit transcriptions

Buddhist translators into Chinese used two devices: 音譯, spelling a Sanskrit word by sound, and 意譯, translating its meaning. Only the first is transcription. A 27,110-entry terminology dictionary mixes them without marking which is which, and the two are not separable by inspection.

Two approaches failed before one worked. Character-set membership admits 福慧 for *puṇya*, which is a calque. Exclusivity — treating a character as phonetic only if it never appears in a translation — collapsed the corpus to 67 pairs. What worked was phonetic verification: score each candidate against the Later Han reading of its characters and keep only those whose sound actually matches. The result is **1,017 verified pairs**, and the 7,480 rejected entries are retained separately so the decision can be audited. The induced 音譯 inventory matches the standard akṣara table, which is an external check that the filter found something real.

3.4 Lexicons 1 and 2 — Kāṣṣarî and the *Codex Cumanicus*

The tests in §6 need something the three corpora are not: dictionaries of Turkic itself, from near the period, against which a reading can be looked up. No downloadable dataset of either was found, so both were extracted.

Kāṣṣarî Mahmud's *Dīwān Lughāt al-Turk* (1072–77) is the earliest dictionary of any Turkic language.¹¹ The Türk Dil Kurumu index volume has a machine-readable text layer; parsing it yields **6,820 headwords** with Kāṣṣarî's Turkish glosses and his volume- and page references.

The *Codex Cumanicus* (c. 1300) records Kipchak Turkic, compiled by Italian merchants and German missionaries — outsiders writing down a language they did not speak, which is the position the Chinese scribes were in. Géza Kuun's 1880 edition prints a Cumano-Latin vocabulary; parsing it yields **2,110 headwords** with Latin glosses.¹³ The Persian and German vocabularies that follow in the same volume were deliberately skipped: the Persian one would inject Iranian vocabulary into a Turkic lexicon and inflate every match rate computed from it.

Uppsala's Middle Turkic Database is building digital editions of Middle Turkic sources under CC BY 4.0 and may in time supersede both; at the time of writing its corpus is Middle Karaim and covers neither work.

4 Method

The channel is a probabilistic model of the scribe: for a source sound, what Chinese spelling would he choose? Character-to-sound alignments are learned by expectation–maximisation over the training pairs, without hand-alignment. Emission tables are *reading-anchored* with backoff — full reading, then onset plus rime, then onset, then a global distribution — so a character never seen in training still receives a distribution rather than nothing.

Decoding runs the channel backwards: given the characters, score candidate source words. The binding variable throughout is not corpus size but **observations per unit** — how many times the model has seen each thing it must estimate. That figure, not the number of pairs, is what predicts whether reconstruction works.

Every comparison in this paper is accompanied by a null. For lexicon searches, pseudo-names are built by resampling Later Han syllables from the same names being tested, so they carry the right shape and the right sound inventory and no meaning at all; the identical search is run on them. For language comparisons, corpora are subsampled to equal size before scoring, because otherwise the thinner model wins by being vague. Negative controls — running the method on cases whose answer is already known — are what exposed the failures reported in §6.

5 What the channel recovers

Every figure below is on held-out data. "Exact" means the whole word was reconstructed correctly; "within 2" means no more than two segments were wrong.

TRAINED ON → TESTED ON	EXACT	WITHIN 1	WITHIN 2
Mongolian → Mongolian (unseen chapters)	36.9%	66.3%	87.5%
Turkic → Turkic	30.6%	68.2%	83.5%
Sanskrit → Sanskrit	25.0%	57.9%	77.0%
Sanskrit → <i>Turkic</i> (wrong training language)	3.1%	6.9%	27.5%

TABLE 1 Each row is one experiment. Row 1 trains on some chapters of the *Secret History* and tests on chapters held back entirely — a document-level split, so no word from a test chapter influenced training. Rows 2 and 3 do the same within the Turkic and Sanskrit corpora, but those are glossaries without document structure, so the held-out portion is a random sample of pairs. Row 4 is the diagnostic: the channel is trained entirely on Sanskrit and asked to read Turkic. A model trained on Buddhist transcriptions barely functions on Turkic ones — 3.1% against 30.6% — which is the clearest evidence that the channel learns something specific to the language it was trained on rather than a generic Chinese-to-anything mapping.

CORPUS	PAIRS	OBSERVATIONS PER CHARACTER
Mongolian	9,329	~60
Sanskrit	1,017	~12
Turkic	594	~7
Xiongnu names	39 items	1.4

TABLE 2 The Xiongnu material is 112 character-tokens over 81 distinct characters. This is the number that governs everything in §6: the method does not fail on Xiongnu because Xiongnu is mysterious, but because 1.4 observations per character is not enough to estimate anything. The comparison is with 60 for the corpus where the same method reaches 36.9% exact.

6 What the evidence cannot support

Two claims commonly made from this kind of material do not hold up, and establishing that carefully is the most useful output of the work.

6.1 Determining which language a name belongs to

The obvious application is settling whether the Xiongnu ruling house spoke Turkic, Yeniseian, Mongolic or something else. The comparison was built properly: score how well each candidate language explains the transcription, equalise corpus sizes, compare by Bayes factor.

It failed its own controls. Given Ming Uyghur transcriptions whose language is known with certainty, the method identified Turkic correctly in 69% of trials against a 50% chance baseline, and in 2 of 24 experimental cells it fell *below* chance. A diagnostic showed why: the phonotactic priors available for these families are close enough to each other that the comparison is largely measuring the reference data rather than the names. A procedure that cannot reliably identify a language it has been given cannot be trusted to identify one nobody has.

6.2 Reconstructing the Xiongnu names in full

At 1.4 observations per character the decoder returns candidates, but they are not constrained by evidence. An intermediate version of this work gated its output on in-domain accuracy and would have printed confident readings — 撑犁 as *ara*, *kara*, *tara* — where true out-of-domain reliability was 3.1%. Re-gating on out-of-domain accuracy removed them. This was the closest the project came to publishing a wrong answer, and the correction is recorded here because the failure mode is general: a decoder always returns its best candidate, and best is not the same as supported.

6.3 Matching readings to dictionary words

There is an obvious next step, and it is the one every reader wants to take. Once 稽粥 is written out as *kei çuk*, it is hard not to hear Turkish *küçük* "small, younger". The method is old: look the reading up in a dictionary and see what it hits.

The trouble is that it always works. Modern Turkish has some twenty thousand words in common use, and a target with three consonants has hundreds of near neighbours. The question is never "is there a match" but "is there more of a match than a string that cannot mean anything would get?" Seven lexicons were built and each was given its own null.

The modern side of the comparison uses open spelling dictionaries rather than a published dictionary's headword list, because a spelling dictionary can be counted. Turkish is the *hunspell-tr* word list, whose 292,000 forms are cut to 21,721 by keeping only words with a wordfreq Zipf score of at least 2.5 — that is, words a speaker actually uses.^{24,22} Kazakh is the *myspell-kk* list, 53,182

forms, romanised on load by a fixed Cyrillic-to-Latin table.¹² The frequency filter matters: without it the Turkish list is dominated by inflected and rare forms, and every target finds a match. The historical side is described in §3.4.

LEXICON	ENTRIES	PSEUDO-NAMES FINDING A MATCH	MEDIAN SCORE FOR A PSEUDO-NAME	NAMES BEATING CHANCE, $P \leq 0.05$
Turkish, common use	21,721	231 / 250	0.156	2 of 36
Kazakh	53,182	239 / 250	0.183	0 of 36
Union — Turkish + Kazakh	73,406	236 / 250	0.117	2 of 36
Turkish, loan-shaped words removed	14,876	233 / 250	0.200	2 of 36
Intersection — in both languages	1,497	219 / 250	0.329	3 of 35
<i>Dīwān Lughāt al-Turk</i> , 1072–77	6,337	276 / 300	0.200	3 of 36
<i>Codex Cumanicus</i> , c. 1300	2,067	275 / 300	0.242	3 of 36

TABLE 3 A lower score is a closer match, so a *lower* median in column four means the lexicon is noisier — meaningless strings are finding better matches in it. Read that column as the price of admission. Two results are worth separating out. **Pooling two dictionaries makes matters worse:** the union finds more matches for the real names and more for the meaningless ones, and the noise floor falls from 0.156 to 0.117 while nothing else moves. A larger haystack does not help you find a needle. What helps is the opposite operation — keeping only words present in *both* languages lifts the floor to 0.329, because a word must survive two independent borrowing histories to qualify. That cure has its own disease: the intersection is over identical spellings, and cognates diverge, so Turkish *tanrı* and Kazakh *tāñir* both fall out while *domino* and *kauçuk*, borrowed into both recently, survive. It enriches for shared modern loans. Across all seven lexicons, between none and three names in thirty-six clear $p \leq 0.05$, which at a one-in-twenty threshold is what thirty-six coin flips give.

Two cases can be checked against a known answer, and the older lexicons do better on both.

NAME	READING	LEXICON	RANK	CANDIDATE	GLOSS IN THAT DICTIONARY
撐犁 chengli	ḍaŋ lei	Modern Turkish	—	not returned	<i>dengeli</i> "balanced" came first
		<i>Dīwān Lughāt al-Turk</i>	4th	tengri	<i>gök, sema</i> — "sky, heaven"
		<i>Codex Cumanicus</i>	2nd	tengri	<i>deus</i> — "God"
頭曼 Touman	do man ^c	Modern Turkish	—	duman	"smoke"; <i>tümen</i> not in the top three
		<i>Dīwān Lughāt al-Turk</i>	3rd	tümen	<i>pek çok</i> — "very many"
		<i>Codex Cumanicus</i>	2nd	touman	<i>nebula</i> — "mist"; <i>tümen</i> absent from this lexicon

TABLE 4 The *Hanshu* glosses 撐犁 as 天 "heaven". Kāšgarî in Baghdad and the compilers of the *Codex* in the Crimea, working a thousand years later with no knowledge of the Chinese record, gloss *tengri* as "sky, heaven" and as *deus*. The attestations are independent and the search connects them — which it fails to do against a modern dictionary, and does better the older and smaller the lexicon becomes. The second name repays attention because it cuts against the accepted etymology: *tümen* "ten thousand" is absent from the Cuman vocabulary entirely, and in every lexicon holding both, the search prefers a different lexeme — Turkish *duman*, Kāšgarî's *tuman*, Cuman *touman*, all "smoke, fog, mist", one word across three lexicons and a thousand years. The difference is a vowel, and §7 puts vowel recovery at 50%, so this settles nothing; it is recorded because the alternative is never raised.

A caution about the strongest datum. The match between *ᠳᠠᠩ lei* and *tengri* is the most quoted evidence for a Turkic Xiongnu. The inference only works if *tengri* is Turkic to begin with. The word is not confined to Turkic: Mongolic has *tenger* "sky", Turkic *tengri* survives as modern Turkish *Tanrı*, and Georg has argued that the Turko-Mongolic word is built on Proto-Yeniseian **təŋgVr-* "high" with a Turkic suffix, making it a borrowing rather than an inheritance.^{8,27} That derivation is one argument among several and need not be accepted; what holds either way is that a word attested across Turkic, Mongolic and — on Georg's account — Yeniseian cannot discriminate between them. 撑犁 shows that the Xiongnu used, for "heaven", a word also found in Turkic and Mongolic. It does not show which of those languages they spoke.

7 Answering the 1974 question

The Xiongnu names survive in Han-period sources; the Turkic training corpus is Ming, twelve hundred years later. Whether anything can be carried across that gap is measurable directly: take characters whose readings are known in both layers, predict the earlier reading from the later one, and score by component.

PART OF THE SYLLABLE	RECOVERED CORRECTLY MING → LATER HAN	95% CONFIDENCE INTERVAL	CHANCE BASELINE — BEST BLIND GUESS	LIFT — POINTS GAINED OVER THAT GUESS	CLASS OF SOUND LAW ¹⁴
Nasal ending (-m, -n, -ng)	99.2%	95.4 — 99.9	35.4%	+63.8	deterministic
Initial consonant	79.3%	72.4 — 84.8	9.1%	+70.1	strong probabilistic
Vowel	50.0%	42.4 — 57.6	15.2%	+34.8	weak probabilistic
Stop ending (-p, -t, -k)	0.0%	0.0 — 7.9	35.4%	-35.4	no law

TABLE 5 $n = 164$ characters (nasal ending 118, stop ending 45). "Recovered correctly" is leave-one-out majority prediction of the earlier reading from the later; intervals are Wilson score intervals; the chance baseline is the unconditional majority class for that part — the score obtained by always guessing the commonest value. **Lift is column two minus column four:** how many percentage points knowing the later reading actually buys. Do not read an aggregate coda figure: blending nasals and stops gives a flattering 87% that means nothing, because the sample is 72% nasal-ending characters.

This is the reviewer's question, answered. Three of the four components classify cleanly. Nasal endings, at 117 of 118, are deterministic for practical purposes. Initial consonants, at 79.3% with the interval nowhere near one half, are a strong probabilistic law. Stop endings are not a law in any sense: not one of 45 survived, and the interval tops out at 7.9%.

The fourth exposes a flaw in the rule itself. Vowels land at **50.0%** — 82 of 164, on the boundary to the decimal — and the confidence interval, 42.4 to 57.6, straddles it. By the reviewer's criterion the vowel cannot be classified, and no larger sample would help, because the estimate is not uncertain; it is genuinely at one half.

But one half is the right threshold only when there are two outcomes. A vowel is a choice among many, and the honest comparison is against what you would get knowing nothing — the majority class, here 15.2%. Against that, vowels carry **+34.8 points** of real information: a weak probabilistic law, not a coin flip. Stop endings, by the same measure, fall *below* their baseline: knowing the Ming reading makes you worse off than guessing, because Old Mandarin merged -p, -t and -k into nothing and the merger is not recoverable.

So the fifty-year-old question has a shape the fifty-year-old debate could not see. The threshold is not a constant. It is the baseline of the component in question, and it moves with how many outcomes that component has. That correction costs nothing to state and could not have been stated without counting.

8 Three measurements of scribal practice

8.1 Is a Chinese m- ever a foreign b-?

Reading 冒頓 as *Bayatur* — the most repeated etymology in the field — requires the m- to stand in for a b-. Proto-Turkic has no initial *m-: the m- words in Old Turkic are loans, or come from *b- assimilating to a nasal immediately after the vowel — **bān* → *men*, **buŋ* → *muŋ*.^{4,7} So on the Turkic reading the substitution is load-bearing. Each corpus is a set of transcriptions whose source word is known, which makes the question a counting exercise.

CORPUS	SOURCE WORDS BEGINNING B-	WRITTEN WITH A CHINESE M- CHARACTER
Mongolian — <i>Secret History</i>	738	0
Turkic — Ligeti glossaries	78	0
Sanskrit — Later Han, verified	22	3
Total	838	3 (0.4%)

TABLE 6 The reverse direction is as clean: Chinese **m**- characters render a source **m**- in 303 of 305 Mongolian cases, 90 of 97 Sanskrit and 14 of 17 Turkic. The decisive detail is that scribes had a choice: Later Han had a voiced **b**- series and used it freely — 239 of the Mongolian **b**- words took a **b**- character and 492 a **p**-. Whoever wrote 冒 had the alternative available and did not take it. The economical reading is that he wrote an **m** because he heard one — which leaves the standard etymology needing a substitution that occurs 3 times in 838 opportunities. Combined with the absent **-r**, that is two independent lines against it, neither depending on the channel model.

8.2 How was a source coda spelled?

A related question, and the one §9 turns on. When a source syllable ended in a consonant, did the scribe use a character ending in that consonant, or an open character with a vowel after it? The Sanskrit corpus answers it for the right period: for the 770 pairs aligning one character to one syllable, each Sanskrit syllable is matched to its character and the coda of each recorded.

SOURCE SYLLABLE ENDS	N	CLOSED WITH THE SAME CODA	WRITTEN WITH AN OPEN CHARACTER
-n	93	83%	13%
-ŋ	22	73%	9%
-t	53	53%	28%
-k	21	48%	38%
-p	10	20%	20%
All stops	84	48%	30%

TABLE 7 Roughly a third of stop-final source syllables were written with an open character, against 13% and 9% for nasals. An open spelling after a stop is ordinary, not exceptional. Note the shape of the asymmetry: nasals are handled robustly and stops loosely — the same split Table 5 found surviving the period gap, appearing here in what the scribe chose to write rather than in what survived. Two independent measurements of one underlying fact.

8.3 When a consonant is written twice, was it said twice?

A scribe could spell one source consonant across a syllable boundary: the first character ends in **-t**, the second begins with **t**-. Whether that doubling records a geminate or is a convention of the script changes how many consonants a transcription actually claims, and therefore how a name may be read. It is answerable on the Sanskrit corpus, where 772 pairs align one character to one syllable.

Taking every junction at which a written stop or nasal coda is followed by a same-place onset, and asking whether the Sanskrit form carries that consonant once or twice:

JUNCTIONS WHERE A WRITTEN CODA IS FOLLOWED BY AN ONSET OF THE SAME PLACE	THE SOURCE WORD HAS THAT CONSONANT ONLY ONCE	THE SOURCE WORD HAS IT TWICE
140	121 (86%)	19 (14%)

TABLE 8 The double spelling is a convention, not a geminate: 釋迦 for *śākya*, 怛他 for *tathā*, 富單 for *pūtana*. A sequence such as 骨都 *kuət ta* therefore transcribes a source *kuta* with one **t**, not *kutta*. The count is conservative: "twice" is scored whenever the doubled letter appears anywhere in the Sanskrit form, so a geminate elsewhere in the word counts against the merger, and 86% is a floor rather than a ceiling. Like §8.2, this is a measurement that settles an objection which would otherwise be argued from intuition — a reader who assumes two written consonants mean two spoken ones will reject readings that the corpus permits.

9 A worked example: what a proposal must survive

Section 6.3 shows that reading a transcription against a dictionary is, on its own, at chance. That is not an argument that no such reading can ever be supported — it is an argument that support must come from somewhere other than the resemblance. One case in this material meets that standard, and it is set out here as a worked example rather than as a result about the Xiongnu.

From 呼韓邪 onward, every chanyu title ends in 若鞮, Later Han *ńak te*. The *Hou Hanshu* states that the element was added to the title from that reign and glosses it 孝 "filial". This is the only Xiongnu title element the Chinese sources translate — which means the meaning is *recorded*, not inferred from the sound.

Four things then have to hold, and each is checkable independently of the others.

- **The sense must be attested.** Searching the two lexicons of §3.4 returns *Inak*, glossed *fidelis* "faithful", in the *Codex Cumanicus*.¹³ It is absent from Kāṣṣarī, so the word belongs to the Kipchak tradition. Old Turkic *inaq* "trusted one, confidant" is generally derived, with *inan-* "to trust", from a root *ina-* "to rely on". The sense the proposal needs is documented in a medieval glossary, and it matches 孝 independently.
- **The morphology must be licensed.** A reading *inaq-ti* is not available: *-ti* is not a noun-forming suffix, and the lexicons show it rather than a grammar having to assert it. Of the 31 Cuman headwords ending in *-ti*, almost all are glossed with Latin perfects — *Ayti: dixit*, *Azti: transgressus est*, *Öwtiti: deterisit* — the definite past ending; the remainder are numerals or Arabic loans. Kāṣṣarī's index has six, none containing a suffix. A title cannot be a past-tense verb. Old Turkic does have a suffix that fits: *-t*, a plural used specifically with *titles* — *tegin* → *tegit*, *tarqan* → *tarqat*.⁷ On that reading the element is **Inakt**, "the faithful ones", and 若鞮 is not one ruler's epithet but a title borne by a line — which is exactly how the Chinese sources use it.
- **Objections to the spelling must be tested, not asserted.** Against a final *-t* it can be objected that Later Han had *-t* final syllables and a scribe would have used one rather than the open 鞮 *te*. That is a claim about scribal habit, and Table 7 counts it: 30% of stop-final source syllables were written with an open character. The objection does not hold.
- **What remains unexplained must be stated.** If the source was *inakt*, the opening *t-* is not written, and 若鞮 has no room for it. That objection stands.

Three of four hold, with the fourth named. That is the standard this paper asks of a proposal, and it is deliberately higher than resemblance: the sense is attested in a source contemporary with neither party to the comparison, the morphology is licensed by a suffix with the right distribution, and the one objection that could be tested was tested and failed.

A further 32 candidate readings of Xiongnu names were produced in the course of this work. None of them meets this standard — by §6.3 they are at the chance rate — and they are therefore **not presented here as findings**. They are included as **Supplementary Material S1** and released in the repository as [author_proposals.csv](#), dated and labelled, because readers will run this search in their heads regardless and it is better that the output is written down than imagined.²⁵

S1 is supplied for the record, not for review. It has the status of a laboratory notebook that happens to be legible: each row gives the Chinese form, the Later Han reading, the author's rendering, the sense proposed for it, and — where one exists — the historical or onomastic parallel that suggested it. Those parallels are drawn mostly from later Turkic naming practice, where a ruler's epithet is a common noun (Bayezid *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed *Avcı* "the hunter"), and they are offered as the kind of thing that would have to be argued rather than as an argument. A reader who accepts none of them loses nothing from the paper; the measurements in §§5–8 do not rest on any of them. The one reading that is argued here, *Inakt*, is in S1 too, marked as the exception, so that the file can be read as a single list with the standard visible in it. S1 also contains a worked demonstration, at §S1.5, of why the decoder's own output must not be quoted in support of an individual reading — the concrete form of the warning in §6.2.

9.1 A second case, which is not ours to propose

The same standard can be applied to a comparison somebody else has already made, and that is the more useful exercise, because the reading is then not ours to want. The Chinese sources record 徑路 *keŋ^c la^c* as a Xiongnu blade — the 徑路刀, used in oath-taking. A comparison with Turkic *kıŋrak* is already in the literature, where it is described as speculative. Our corpus building recorded it during the assembly of the Xiongnu lexicon, flagged unverified. It can now be tested.

- **The sense is attested, and attested of the right object.** Kāšgarî glosses *kıŋrak* as a large knife resembling a cleaver, for cutting meat and dough. The Chinese side independently says the referent is a blade. This is a stronger kind of agreement than a sound match: two sources with no contact agree about *what the thing is*.
- **The consonant frame is exact.** 徑 gives the velar onset with its -ŋ coda written; 路 gives a Chinese ɭ- for a source *r*-, which §8.1's companion count shows is the ordinary direction — a source *r*- was written ɭ- in 12 of 14 attestations.
- **The one spelling objection is measured, not argued.** The Turkic word ends in -k and the transcription does not write it. Table 7 gives the rate at which a stop-final source syllable was spelled with an open character: 30%.
- **What remains unexplained is stated.** The vowels are not recoverable at the 50% of §7. And a second word of the identical shape, *kaŋrak* "palate", fits the transcription exactly as well; it is excluded only by the referent, which is precisely the evidence a pure sound match does not have.

So an existing speculative comparison survives the tests that 32 of our own readings fail. That asymmetry is the point of this section. The method here cannot generate supported readings at this data density, but it can adjudicate ones already on the table — and adjudicating someone else's proposal is a use of the channel that its accuracy figures actually license.

The attribution of the *kıŋrak* comparison is recorded in our lexicon file as "cited" without a name, and must be traced to its original publication before this section can stand as written. It is included here because the test result does not depend on who first proposed it.

10 Where the method does work

The constraint identified in §5 is data density, not the difficulty of the language. That suggests where to go next. Göktürk-era material has what the Xiongnu material lacks: a source language that is not in dispute, an independent script recording it, and enough of it to train on.

CHINESE, AS WRITTEN IN THE TANG HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	INDEPENDENTLY IDENTIFIED AS, IN ESTABLISHED SCHOLARSHIP
葉護	jap ɣua ^c	yap ǵua	<i>yabghu</i> , a princely title
可汗	kʰai ^B gan ^c	kai gan	<i>qayan</i>
特勤	dək giən	dık gıın	<i>tigin</i> , prince
突厥	tʰuət kyat	tuıt kyat	<i>Türk</i>
阿史那	ʔa ɕiə ^B na	ʼa ɕiı na	<i>Ashina</i> , the ruling clan
黠戛斯	get kət sie	get ket sie	<i>Qırqız</i> (Kyrgyz)

TABLE 9 Neither middle column is model output. Column two is Schuessler's published Later Han reconstruction applied to Tang-era words — deliberately the wrong period, which is the point of the exercise. The last column is not an output of this work either: each identification is long settled on evidence independent of the Chinese transcriptions, the Orkhon inscriptions giving *qayan*, *tigin*, *yabghu* and *Türk* directly in Old Turkic script, several corroborated again in Sogdian, Arabic and Byzantine

sources.^{7,9} They function here as ground truth. That 葉護 → *yap ġua* is readable against *yabghu* despite a period mismatch of four centuries measures the conservatism of Chinese scribal practice — not linguistic continuity between the Xiongnu and the Göktürks, on which this material bears not at all.

Chinese records name a great many steppe peoples, and they fall into three groups for these purposes. **Tractable**: those recorded in the Tang histories whose language is known — 鐵勒 *Tiele*, 回紇 *Uyghur*, 葛邏祿 *Qarluq*, 突騎施 *Türgesh*, 黠戛斯 *Kyrgyz*, alongside the Göktürk material. These have transcriptions, a securely identified source language and published readings for the right layer; they are training data. **Readable but uninformative**: groups such as 丁零 *Dingling* and the Avars, whose Chinese readings can be recovered but whose language is unknown, so a reading cannot be checked. **Out of reach**: the Xiongnu themselves, and the Pechenegs, who are barely in the Chinese record at all. The distinction is not how interesting the group is but whether there is anything to validate against.

11 Limitations

- **No independent philological review.** The computational method and its evaluation are the author's to defend. The philology is not the author's work and has not been checked for this paper: the Later Han readings are Schuessler's, the etymologies discussed are from the published literature, and the identifications in Table 9 are the established consensus. A specialist reading would strengthen the paper and might revise judgements in §6 and §9.
- **Corpus validation is internal.** The extraction rules are documented and internally consistent — 93.4% length agreement for Mongolian, an induced akṣara inventory matching the standard one for Sanskrit — but no external reviewer has checked a sample against the sources.
- **The Turkic corpus is Ming-era**, roughly 1,200 years later than the Xiongnu material. §7 measures what that costs rather than assuming it away.
- **Reconstructions are inferences.** Schuessler, Pulleyblank and Ning are three scholars with three notations; some apparent disagreement between them is notational rather than substantive.
- **Names travel.** Even a perfect reconstruction would not settle ethnicity. Turkic tribes bore Iranian names and Mongol rulers used Turkic titles; the caution in §6.3 about *tengri* is a specific case of this general problem.
- **Reproducibility has a version dependency.** The headword count for the Kāṣṣgarī lexicon differs between machines — 6,820 against 6,880 — according to the version of the PDF text-extraction library. It shifts the noise floor by 0.008 and changes no rank or conclusion, but it is the kind of thing that stays invisible until someone else runs the code.

12 Data and code availability

Everything described here is packaged as one repository, so that every number and table in this paper can be regenerated from the raw inputs by running the numbered scripts in order. The repository is the durable output; this paper describes it.²⁵

```
chinese-transcription-channel/
├─ README.md           what each output is, and how to regenerate it
├─ LICENSE             code MIT
├─ DATA_LICENSE.md    the three-way split described below
├─ data/derived/       the corpora, the lexicons, and every measurement
├─ src/                steps 1–22 plus lookup_lexicons.py, each rerunnable alone
├─ reports/            one summary per step, with the numbers quoted above
└─ docs/               design notes, including approaches that were rejected
```

Supplementary Material S1. *Candidate readings of the Xiongnu record* accompanies this submission and is the same content as [data/derived/author_proposals.csv](#) in the repository. It is supplied for the record under §9 and is not offered as a result of this paper.

Released, CC BY 4.0. The Mongolian and Sanskrit corpora, the Cuman lexicon (Kuun's 1880 edition being public domain), the induced akṣara table, the Xiongnu record of Appendix A, and every measurement file behind Tables 1–9.

Withheld, with the extraction script released instead. Four derived files inherit restrictions from their sources: the Turkic pairs, from Ligeti's glossaries (© Akadémiai Kiadó); the Kāṣṣgarī lexicon, from a modern Türk Dil Kurumu edition; and two files built on that lexicon — the Old Turkic echo results and the candidate-space listing, the latter also quoting Clauson's *Etymological Dictionary* verbatim in a context column. Anyone holding the sources can regenerate any of them byte-for-byte by running the numbered script, which keeps the paper's numbers checkable without redistributing the source. No permission is required for this arrangement and none has been sought.

Not in the repository. Third-party source texts and datasets — dictionaries, scanned volumes, UniHan, the DILA authority files — are inputs, not outputs. Where a script needs one, its docstring says which file and where to obtain it.

A Appendix: the Xiongnu record

All thirty-nine names, titles and lexical items in the Chinese histories, with the Later Han reading of each character and a Turkish-orthography rendering for readers who do not read phonetic notation. The Later Han column is published scholarship, not model output; the contribution here is the assembly — the names collected in one place with the readings joined to them character by character. Superscript letters mark Han tone categories. 若鞮 is listed once, shaded, rather than repeated in the six titles that carry it.

A.1 Rulers and the ruling clan

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
頭曼 Touman	do man ^c	do man	209 BCE
冒頓 Modu / Modun	mæk tuən ^c	mık tuın	209–174 BCE
稽粥 Jizhou / Jiyu	k ^h ei ^B tśuk	kei çuk	174–160 BCE
軍臣 Junchen	kun dźin	kun cin	160–126 BCE
伊稚斜 Yizhixie	ʔi ɖi ^c ja	’i di ya	126–114 BCE
烏維 Wuwei	ʔa wi	’a vi	114–105 BCE
詹師廬 Zhanshilu	tśam ʃi lia	çam ʃi lia	105–102 BCE
呼犁湖 Xulihu / Goulihu	huo lei ga	huo lei ga	102–101 BCE
且鞮侯 Qiedihou	tsa ^B te go	tsa te go	101–96 BCE

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
狐鹿姑 Hulugu	ɣua lok ka	ğua lok ka	96–85 BCE
壺衍鞬 Huyandi	ga jan ^B te	ga yan te	85–68 BCE
虛間權渠 Xulüquanqu	k ^h ia lia gyan gia	kia lia gyan gia	68–60 BCE
握衍胸鞬 Woyanqudi	ʔok jan ^B guo te	’ok yan guo te	60–58 BCE
呼韓邪 Huhanye	ha gan ja	ha gan ya	58–31 BCE
郅支 Zhizhi	t́sit t́se	çit çe	56–36 BCE
若鞬 Ruodi	ńak te	nyak te	title element
復株累若鞬 Fuzhulei Ruodi	buk ʦio lui ńak te	buk tio lui nyak te	31–20 BCE
搜諧若鞬 Souxie Ruodi	so ɣei ńak te	so gei nyak te	20–12 BCE
車牙若鞬 Cheya Ruodi	t́sʰa ɲa ńak te	ça nga nyak te	12–8 BCE
烏珠留若鞬 Wuzhuliu Ruodi	ʔa t́so liu ńak te	’a ço liu nyak te	8 BCE–13 CE
烏累若鞬 Wulei Ruodi	ʔa lui ńak te	’a lui nyak te	13 CE–18 CE
呼都而尸道皋若鞬 Huduershi Daogao Ruodi	ha ta ńə sí dou ^B kou ńak te	ha ta nyı şı dou kou nyak te	18 CE–46 CE
攣鞬 Luandi	lyan te	lyan te	—
虛連題 Xulianti	k ^h ia lian ^B de	kia lian de	—

Twenty-one *chanyu* from Touman to Punu, plus both forms of the ruling clan name — 攣鞬 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*.

A.2 Titles and lexical items

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
單于 chanyu	d́zan wa	can va	title
撐犁孤塗單于 chengli gutu chanyu	ɖaŋ lei kua da d́zan wa	dang lei kua da can va	title

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
撐犁 chengli	ḍaŋ lei	dang lei	lexical
孤塗 gutu	kua da	kua da	lexical
閼氏 yanzhi / ezhi	ʔat dʒe ^B	ʼat ce	title
屠耆 tuqi	da ɣi	da ɣɪ	title
谷蠡 luli / guli	kok le ^B	kok le	title
當戶 danghu	taŋ ga ^B	tang ga	title
骨都侯 guduhou	kuət ta go	kuɪt ta go	title
且渠 juqu	tʂa ^B ɣia	tʂa ɣɪa	title
徑路 jinglu	keŋ ^c la ^c	keng la	lexical
服匿 funi	buk ɳik	buk nɪk	lexical
甌脫 outuo	ʔo tʰuat	ʼo tuat	lexical
胡祿 hulu	go lok	go lok	lexical
匈奴 Xiongnu	huoŋ na	huong na	ethnonym
羯 Jie	kɪat	kɪat	ethnonym

The lexical items matter most: they are the only Xiongnu words the Chinese sources gloss. 撐犁 is glossed "heaven" and 孤塗 "son", which is why that pair has carried so much of the argument. Note that the Yeniseian proposal for 孤塗 is *(h)ala* "son" and the Turkic one *qut* "fortune"; the Chinese gloss 子 "son" fits the former and not the latter.^{26,27}

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